Regularized Market Timing under Trading Costs: A Reproducible Study of Daily SPY Returns

DSA5205 Project 2 | November 2026

Abstract

Can a small regularized return model improve daily equity timing once its forecasts are translated into executable, cost-bearing positions? We study SPY using five trailing trend and volatility features, monthly Ridge refits, and a fixed 1,260-observation window. Four regularization strengths are compared on 2019–2021 validation data; the selected model is then evaluated on 2022–2025 without further strategy selection. Features, training labels, executions, and dividend entitlements are aligned explicitly in time. At a single-side cost of 5 basis points, the strategy achieves 6.54% compound annual growth and a 0.481 Sharpe ratio, versus 11.18% and 0.670 for buy-and-hold. Its maximum drawdown is smaller, but equity exposure is only 69.36%. Gross Sharpe parity does not survive the strategy's 246 units of one-way turnover. Prediction R^2 against a past-only historical mean is -0.00541. The finding is descriptive rather than a significance claim: regularization and leakage controls make the experiment credible, but do not establish a net timing advantage. Source code, frozen artifacts, accounting derivations, and integrity checks accompany the study.

1 Introduction

Financial prediction is useful only insofar as it supports a feasible investment decision. A forecast can appear attractive under squared-error loss yet be too small to trade, too unstable to hold, or too expensive to implement. Conversely, lower portfolio volatility can result from holding less equity rather than from identifying favorable return intervals. Evaluating these possibilities requires a chain of evidence from information availability to executable returns, portfolio accounting, and a fair passive alternative.

This paper examines that chain in a deliberately constrained setting. We forecast daily SPY holding returns with five price-based features and Ridge regression. Positions are either fully invested in SPY or fully in cash. A model is refitted monthly on the most recent 1,260 realized outcomes, and its regularization strength is selected from four candidates using a forward validation period. The setting is small enough that every modeling choice, training window, and trade can be inspected. It also provides a useful test of whether implementation discipline is sufficient to turn plausible technical predictors into a net investment advantage.

The economic motivation allows competing effects rather than imposing a guaranteed signal. Recent returns may persist when information or portfolio adjustments diffuse gradually. Very recent changes may reverse after temporary liquidity pressure. Volatility describes uncertainty surrounding these patterns. These mechanisms motivate the chosen lookbacks, but are not causally identified by this experiment. A regularized linear model estimates their joint predictive association while limiting instability from correlated inputs.

The central result is a failure to establish net superiority. At zero cost, the model's annualized Sharpe is approximately 0.678, close to buy-and-hold's 0.672. With a 5-basis-point single-side cost, its Sharpe falls to 0.481, below the benchmark's 0.670. Compound annual growth is 6.54% rather than 11.18%. Maximum drawdown improves from -26.28% to -22.20%, but the strategy is invested during only 69.36% of intervals. The observed risk reduction therefore cannot, by itself, demonstrate timing skill.

Our contribution is an auditable case study, not a new return-prediction algorithm or a claim of statistical alpha. Three aspects distinguish the evidence presented here. First, the forecast target is the opening-to-opening return that follows the decision, including explicit dividend entitlement. Second, eligible training outcomes are filtered by the time at which their holding periods end; a simple row shift is insufficient. Third, we connect the net result to a precise accounting identity: for binary positions and a fixed proportional fee, the ratio of net to gross terminal wealth is determined exactly by the number of charged one-way trades. This makes the cost mechanism transparent instead of attributing all underperformance to an unspecified regime change.

The study preserves the negative finding. No alternative feature set, threshold, estimation window, or model family is selected after examining the final holdout. Additional tables describe the frozen predictions and accounting path; they do not create a second strategy search. The conclusion is correspondingly narrow: this particular small model does not earn a robust net advantage over its investable benchmark under the stated assumptions.

2 Related Work and Scope

Regularization provides a natural response to unstable coefficients in correlated prediction problems. Ridge regression trades additional bias for reduced estimation sensitivity through an ℓ_2 penalty (Hoerl and Kennard, 1970). We use that established estimator as an interpretable baseline. With only five features and substantially more training observations, the experiment is not a high-dimensional interpolation or model-complexity study.

Two assignment-provided examples illustrate a broader distinction between statistical modeling and portfolio evaluation. Chinco et al. (2019) use LASSO to identify short-lived, sparse predictors in one-minute cross-sectional returns. Kelly et al.

(2024) investigate return prediction with many parameters and shrinkage. Their markets, horizons, information sets, and research questions differ materially from ours. The present daily, single-asset experiment neither replicates those papers nor tests their general conclusions about complexity.

A past-only historical mean is an important forecast reference in return prediction (Campbell and Thompson, 2008). We therefore report out-of-sample squared-error performance relative to a rolling historical-mean forecast built from the same mature training sample. The target here is a nominal SPY holding return, not the excess-return target of that literature. The analogy concerns benchmark discipline, not an assertion of identical economic quantities.

Our principal investment benchmark is buy-and-hold SPY. A fixed 20-session moving-average rule supplies a second, simple timing comparison. Neither benchmark is tuned on the holdout. The comparison asks whether the selected model adds value relative to alternatives that an investor could implement at the same dates and under the same execution convention. An additional constant-exposure series is used only for ex post attribution and is explicitly excluded from the set of investable benchmarks.

3 Data and Executable Prediction Target

3.1 Asset, coverage, and cleaning

The asset is SPY, denominated in US dollars. The requested data range is 1 January 2010 through 1 January 2026, with the latter endpoint excluded. Yahoo Finance data are obtained using `yfinance` with `auto_adjust=False`, corporate actions enabled, and automatic repairs disabled (yfinance contributors, 2026). The resulting sample contains 4,024 sessions, from 4 January 2010 through 31 December 2025. Separate raw OHLC, adjusted-close, volume, dividend, and split fields permit independent feature and execution conventions.

Observed dates are checked against the XNYS session calendar, including special closures (exchange_calendars contributors, 2026). There are no missing or extra sessions, duplicate dates, missing fields, or zero-volume observations. The implementation also checks positive finite prices, nonnegative distributions, and valid OHLC ordering. It fails on unsupported data rather than silently filling prices. Extreme observed returns are retained; neither full-sample winsorization nor interpolation is used. Twenty initial observations are excluded for feature warm-up and two terminal observations for unavailable outcomes, leaving 4,002 feature-ready labeled samples across the full history.

SPY has 64 dividend events and no split events in the downloaded interval. The implementation explicitly refuses a nonzero split event pending extended share accounting. Thus the absence of split handling is a stated domain restriction rather than an unobserved modeling error. Raw data remain in a local ignored cache. Provenance records the download options, coverage, package version, retrieval timestamp, and serialized-file digest; published outputs contain derived analysis rather than the full raw price history.

3.2 Information and target alignment

Let t index exchange sessions, and let $\mathcal{F}_t^c$ denote information available after close t. A feature vector x_t and forecast $\widehat{y}_t$ are formed at that time. The position is executed at open $t+1$ and marked through open $t+2$. If O_t is the raw opening price and D_t the distribution with ex-date t, the target is

$$y_t = \frac{O_{t+2} + D_{t+2}}{O_{t+1}} - 1. \tag{1}$$

Here the subscript identifies the signal date, not the date on which the outcome becomes observable. A position purchased at the ex-date open is not entitled to that day's distribution; a position carried into the exit ex-date is entitled. Equation (1) credits the corresponding receivable at the exit open. Actual payment-date cash timing is abstracted away. We do not add cash dividends to an already total-return-adjusted execution price.

The distinction matters for training. At close d, the sample with outcome ending at open d is realized, but a sample ending at open $d+1$ is not. Define its exit session by $e(t) = t+2$. The eligible set is

$$\mathcal{T}_d = \mathrm{tail}_{1260}\{t : t < d,\ e(t) \le d\}. \tag{2}$$

Both the signal-date ordering and label-maturity condition are enforced in code. Training simply on all rows with an earlier feature timestamp would admit an unknown future return.

3.3 Five trailing features

Let A_t be adjusted close and $r_t^A = A_t/A_{t-1} - 1$. The feature vector contains

$$m_t^{(k)} = A_t/A_{t-k} - 1, \quad k \in \{1, 5, 20\}, \tag{3}$$

$$g_t = \frac{A_t}{\frac{1}{20}\sum_{j=0}^{19} A_{t-j}} - 1, \tag{4}$$

$$v_t = \mathrm{sd}_{19}(r_{t-19}^A, \ldots, r_t^A). \tag{5}$$

The notation sd_{19} indicates the sample standard deviation of 20 returns with denominator 19. All inputs are available in $\mathcal{F}_t^c$. The return lookbacks represent short and intermediate trends; g_t measures location relative to a recent average, and v_t describes recent uncertainty.

Vendor adjusted histories may incorporate subsequent corporate actions. These features depend on ratios and returns rather than adjusted price levels. A later common multiplicative adjustment on an earlier lookback window cancels from each feature. Appendix B states and proves this invariance. Arbitrary vendor revisions need not be common factors and remain a reproducibility limitation, addressed by retaining the input digest.

4 Regularized Forecasting and Portfolio Construction

4.1 Past-only standardization and Ridge fitting

At each refit date d, feature means μ_d and population standard deviations s_d are computed exclusively on $\mathcal{T}_d$. Standardized features are $z_{t,d} = (x_t - \mu_d) \oslash s_d$, with the library's constant-feature handling if needed. Ridge solves

$$(\widehat{a}_d, \widehat{\beta}_d) = \arg\min_{a,\beta} \sum_{t \in \mathcal{T}_d} (y_t - a - z_{t,d}^{\top}\beta)^2 + \alpha \|\beta\|_2^2. \tag{6}$$

The intercept is unpenalized. The implementation uses a `StandardScaler/Ridge` pipeline and the deterministic SVD solver (scikit-learn developers, 2026). Its penalty convention is the sum-of-squares objective in Eq. (6), not average squared loss; the numerical scale of α should therefore not be transferred to another normalization without adjustment.

The model is refitted at the first signal and the first decision session of every subsequent calendar month. Between refits, coefficients and scaling remain fixed while x_t changes daily. At least 1,000 eligible observations are required. The actual window contains the most recent 1,260 mature labels, approximately five trading years rather than exactly five calendar years. This gives a controlled amount of estimation history while permitting scheduled adaptation.

Shrinkage is motivated by correlated lookbacks, not by an assumption that any particular predictor must work. A larger penalty limits coefficient magnitudes, but does not eliminate model misspecification or guarantee profitable forecasts. With correlated inputs, individual coefficient signs are conditional descriptions and should not be read as causal estimates of momentum or reversal.

4.2 Forecasts, positions, and baselines

Let $d(t)$ be the most recent refit date. The forecast and target position are

$$\widehat{y}_t = \widehat{a}_{d(t)} + z_{t,d(t)}^{\top}\widehat{\beta}_{d(t)}, \tag{7}$$

$$w_t = \mathbb{I}\{\widehat{y}_t > 0\}. \tag{8}$$

Weights are exactly zero or one. There is no leverage, shorting, borrowing, or intermediate fractional allocation. Cash earns zero. The threshold is fixed at zero before results are inspected. This simple map discards forecast magnitude and can trade on small changes near zero; that limitation is part of the tested design, not corrected after seeing the holdout.

Buy-and-hold uses $w_t = 1$ throughout the common holding interval. The moving-average baseline uses $w_t^{\mathrm{MA}} = \mathbb{I}\{g_t > 0\}$. Both use the same next-open timing, dividend convention, and fee assumptions as Ridge. The 20-session baseline window is fixed rather than selected from the data.

4.3 Transaction costs and wealth

Index the N evaluated holding intervals in chronological order. Set $w_0 = 0$ before the first entry and let $q_i = |w_i - w_{i-1}|$. For single-side cost c, interval growth before terminal liquidation is

$$G_i(c) = (1 - cq_i)(1 + w_i y_i). \tag{9}$$

The final interval is additionally multiplied by $(1 - cw_N)$ to liquidate any remaining position. Fees are fractions of pre-trade equity and are deducted before the interval's market return. This is a stated proportional haircut convention, not a detailed broker commission model. With binary weights, no allocation-drift rebalancing is required between position changes.

The main assumption is $c = 5/10^4$. Sensitivity scenarios set c to zero and $10/10^4$, holding predictions and positions unchanged. Buy-and-hold pays for its initial purchase and terminal sale. Thus its low turnover is an economic property of the benchmark, not an exemption from cost accounting.

Proposition 1 (Exact turnover drag)**.** *For fixed binary weights, fixed $c \in [0, 1)$, and positive gross interval wealth, let $K = \sum_{i=1}^{N} q_i + w_N$. Net and zero-cost terminal wealth satisfy*

$$\frac{V_N(c)}{V_N(0)} = (1 - c)^K. \tag{10}$$

Every charged trade contributes the same multiplicative factor and no unchanged position contributes a fee. The full derivation and numerical verification appear in Appendix C. This identity concerns the frozen path; it is not a model of how trades or liquidity would respond to a different real execution environment.

4.4 Overfitting controls

The experiment fixes one asset, five features, one model family, a 1,260-observation window, monthly refits, and a zero forecast threshold. Only $\alpha \in \{0.1, 1, 10, 100\}$ is selected. Validation evaluates the full trading rule after costs rather than optimizing an unrelated in-sample fit metric. The selected setting is locked before final evaluation. This restriction limits search degrees of freedom, although even a four-candidate validation choice can be optimistic.

The pipeline exports each fitted window, label-maturity boundary, scaler, intercept, and coefficient vector. These records provide an operational check of the information set, beyond a general assurance that the backtest is out of sample. Unit tests perturb future prices and confirm that earlier features, fits, and forecasts do not change.

5 Experimental Protocol

5.1 Chronology and selection

The initial development history is 2010–2018, validation uses holding intervals within 2019–2021, and final testing uses holding intervals within 2022–2025. An interval belongs to

Table 1. Validation-only selection, 2019–2021. All returns include 5 bps single-side costs.

α	CAGR (%)	Net Sharpe	Exposure (%)
0.1	22.33	1.2060	76.72
1	22.33	1.2060	76.72
10	19.15	1.0767	77.38
100	18.64	1.0310	85.71

a reporting period only if its entry is on or after the period start and its exit is on or before the period end. The first holdout signal is therefore formed at the 31 December 2021 close, for entry on 3 January 2022. The final exit is the 31 December 2025 open. No 2021 holding return enters the test merely because its signal date precedes entry.

Each of the four candidates is walked forward through the validation period, with scaling and model parameters fitted only on past eligible observations. The candidate with the highest net validation Sharpe at 5 bps is selected; ties within 10^{-12} choose the larger α. Table 1 gives the complete declared grid. Penalties 0.1 and 1 produce the same validation position path, although their continuous forecasts differ. The tie rule selects $\alpha = 1$.

The selection artifact stores the protocol digest, raw-data digest, validation-table digest, candidate count, and selected penalty. The final-test routine checks those fingerprints and fits only the selected penalty. Repeating the identical pipeline verifies reproducibility; it does not provide a fresh independent holdout. No feature, threshold, window, or model-family change is justified using these test results.

The holdout contains 1,002 return intervals and 49 model fits: an initial fit on the last session of 2021 and scheduled fits at the first decision session of each month in 2022–2025. Earlier holdout outcomes can enter later rolling windows after they mature. That scheduled updating differs from reselecting a strategy after observing its test performance.

The design was fixed before the first market-data download in this implementation on 9 November 2026. It is not an externally preregistered prospective test: these market years were already historical, and researchers may know their broad behavior. The artifact lock documents the implemented sequence, not a stronger guarantee about historical research independence.

5.2 Evaluation measures

For net interval returns $R_i = G_i - 1$, with final liquidation included, define

$$\mathrm{CAGR} = V_N^{252/N} - 1, \tag{11}$$

$$\widehat{\sigma}_{\mathrm{ann}} = \sqrt{252}\ \mathrm{sd}_{N-1}(R_1, \ldots, R_N), \tag{12}$$

$$\widehat{\mathrm{SR}} = \frac{252\,\overline{R}}{\widehat{\sigma}_{\mathrm{ann}}}. \tag{13}$$

The Sharpe reference rate is zero, consistent with the cash assumption. It is not an estimate of excess performance over realized Treasury returns. Arithmetic annualized return is $252\overline{R}$, reported separately from CAGR. Maximum drawdown is the minimum wealth-to-running-peak ratio minus one, with initial wealth one included in the peak. Turnover counts both initial entry and terminal liquidation.

Algorithm 1 Time-respecting selection and evaluation

Require: Trailing features, labeled exit dates, fixed grid $\mathcal{A}$
for $\alpha \in \mathcal{A}$ **do**
  for validation decision close t, in order **do**
    if first signal or new decision month **then**
      Select $\mathcal{T}_t$ by Eq. (2)
      Fit scaler and Ridge on $\mathcal{T}_t$
    end if
    Forecast and set w_t by Eq. (8)
  end for
  Account for costs, entry, and liquidation
end for
Select α^* by validation net Sharpe
Lock protocol, data, and selection digests
Walk forward on holdout using α^* only
Save forecasts, fit audits, wealth, and metrics

Forecast error is assessed against a historical-mean forecast $\bar{y}_{d(t)}$ estimated from the same mature training sample at each refit:

$$R^2_{\mathrm{OOS}} = 1 - \frac{\sum_i (y_i - \widehat{y}_i)^2}{\sum_i (y_i - \bar{y}_{d(i)})^2}. \tag{14}$$

This is neither in-sample explained variance nor a comparison with the holdout mean. Annual return tables use the continuous net portfolio path, without artificial December liquidation and January repurchase.

6 Results

6.1 Net performance does not beat the passive alternative

Table 2 reports the full-period holdout results at 5 bps. Ridge earns 6.54% CAGR and a 0.481 Sharpe, compared with 11.18% and 0.670 for buy-and-hold. Its total net return is 28.63%, versus 52.38% for the passive alternative. It exceeds the fixed moving-average rule on CAGR and Sharpe, but outperforming another timing rule is insufficient to establish economic value when an investable passive benchmark does better.

Figure 2 shows the cumulative paths. Final wealth is approximately 1.286 for Ridge, 1.155 for MA20, and 1.524 for buy-and-hold. A strategy that spends time in cash must avoid enough adverse returns to compensate for forgone positive exposure and extra fees. The positive average return of the underlying during the holdout makes this hurdle economically relevant. The figure also discourages extrapolating from the strategy's stronger validation period or a single favorable test year.

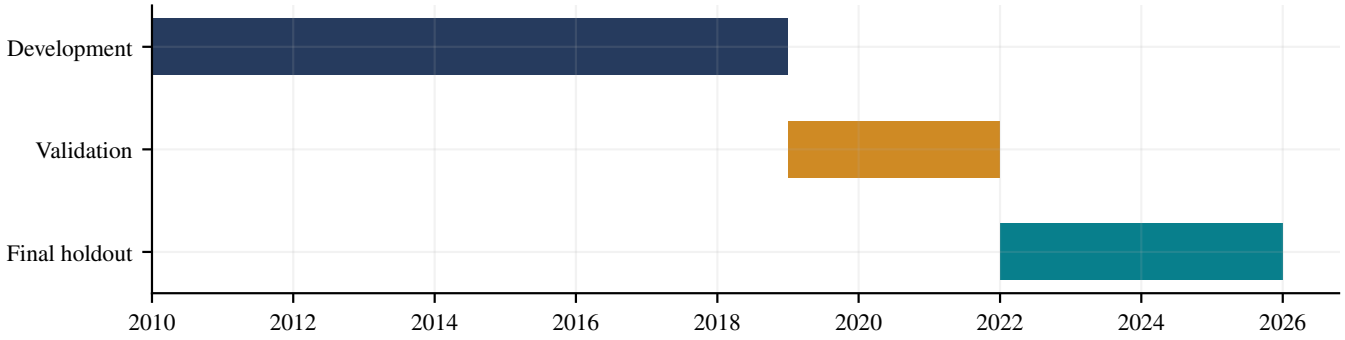


Figure 1. Frozen chronology. Each monthly fit uses only the latest 1,260 outcomes that have matured at the decision close. Validation selects the penalty; the final holdout evaluates only that selection. The split refers to executable holding intervals, not merely feature timestamps.

Table 2. Final holdout, 2022–2025: net performance at 5 bps single-side cost. All strategies share entry and exit dates, dividend treatment, and cost conventions. Returns and volatility are annualized with 252 intervals; total return and exposure are not annualized.

Metric	Ridge	MA20	Buy-and-hold
CAGR (%)	6.54	3.69	11.18
Arithmetic return (%)	7.56	4.22	12.27
Volatility (%)	15.72	10.83	18.31
Sharpe ($r_f = 0$)	0.481	0.389	0.670
Max. drawdown (%)	-22.20	-19.14	-26.28
Total return (%)	28.63	15.52	52.38
Equity exposure (%)	69.36	65.27	100.00
One-way turnover	246	102	2

6.2 Costs erase approximate gross Sharpe parity

The zero-cost Ridge Sharpe is 0.678, compared with 0.672 for buy-and-hold. This approximate numerical parity should not be interpreted as a statistically significant difference. It disappears at the main implementation cost: Ridge's net Sharpe is 0.481, while the benchmark changes little because it trades only twice. Figure 3 varies fees on the same fixed positions. At 10 bps, Ridge's CAGR is 3.29% and Sharpe is 0.284; no new model is selected for any cost scenario.

The mechanism is quantitatively specific. Ridge's annualized turnover is approximately 61.87 units, versus 0.50 for buy-and-hold over this finite entry-to-liquidation sample. From Proposition 1, its net-to-gross terminal wealth ratio is $(1 - 0.0005)^{246} \approx 0.8842$. This is approximately 3.09% annual log-wealth drag. The corresponding benchmark ratio is $(1-0.0005)^2 \approx 0.9990$. Cost sensitivity is therefore linked to a verifiable accounting identity, rather than inferred only from a visual change in the Sharpe curve.

This result does not establish that 5 bps is a universal estimate of SPY opening-auction execution costs. It is a fixed research assumption, with zero and 10 bps included to expose sensitivity. Nor does it imply that a cost-aware threshold would fail. Such a threshold is a different strategy requiring its own selection and evaluation; it is not retrofitted here.

6.3 Risk reduction is accompanied by lower exposure

Ridge's annualized volatility is 15.72%, below the benchmark's 18.31%. Maximum drawdown improves from -26.28% to -22.20%. Figure 4 shows that the strategy still suffers substantial peak-to-trough losses. MA20 reduces volatility further, to 10.83%, but also earns the lowest CAGR. These comparisons demonstrate a risk–growth trade-off rather than a uniform ordering of strategies.

Average Ridge exposure is 69.36%. A constant fractional exposure equal to that realized test average gives an ex post, gross diagnostic with 8.03% CAGR, 12.70% volatility, and 0.672 Sharpe. It uses holdout information and excludes fractional rebalancing costs; it is therefore not an investable comparator. Its role is limited: it illustrates why holding less equity can reduce volatility without establishing selection skill. A genuine risk-matched benchmark would need a pre-test exposure choice or a pre-specified adaptive rule.

6.4 Prediction accuracy supplies no independent rescue

The model's R^2_{OOS} is -0.00541 against the past-only historical mean. Model MSE is 0.00013365, versus 0.00013293 for the baseline. Thus the model does not improve squared-error prediction in the final period. Directional accuracy is 51.20%, a descriptive statistic that ignores the unconditional fraction of positive intervals, return magnitudes, and costs.

The sign-based portfolio rule is especially sensitive to the

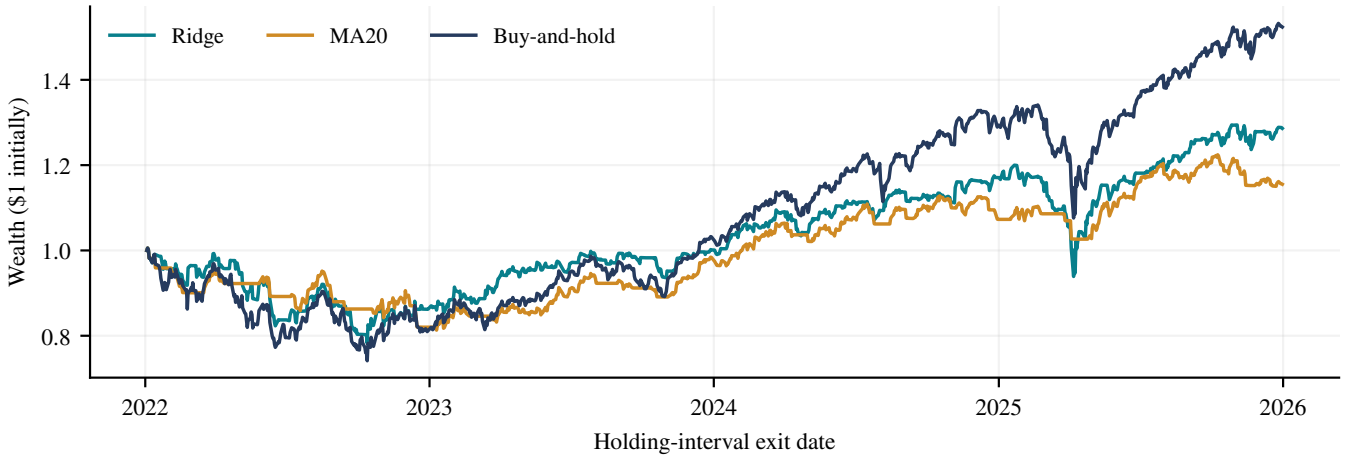


Figure 2. Cumulative net wealth. Every strategy begins with one unit of capital and pays entry and terminal-liquidation fees. Ridge's reduced participation limits losses in some intervals but also sacrifices gains. The plotted trajectory is the frozen holdout path, not a selected subperiod.

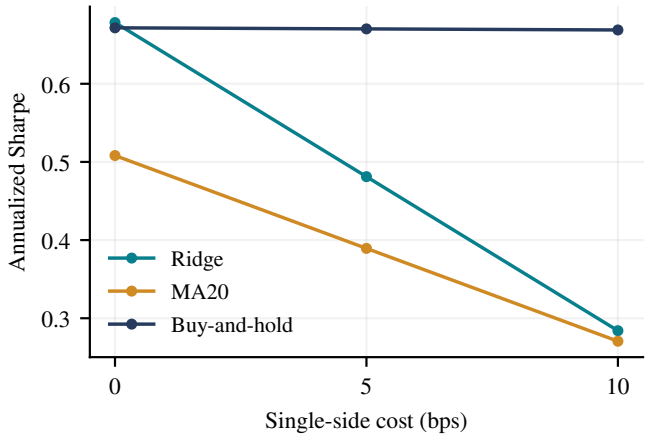


Figure 3. Cost sensitivity of fixed positions. Ridge generates 246 charged one-way trade units, MA20 102, and buy-and-hold 2. The cost scenarios reprice the same experiment.

distinction between prediction and action. Magnitude information is discarded after zero-thresholding: a tiny positive forecast produces the same full position as a larger forecast. Small changes around zero can cause a full turnover event. Forecast scatter, the distribution of scores, a confusion matrix, and realized returns conditional on the frozen decisions appear in Appendix D. Those tables diagnose the same path; they are not used to select another threshold.

6.5 The full-period finding is not a single-year effect

Ridge loses less than buy-and-hold in 2022, but trails it in 2023, 2024, and 2025 (Table 3). The improvement in one down-market year is not sufficient to offset lower growth in subsequent years. Annual Sharpe estimates are short-sample descriptions, not reliable long-run parameters. We do not switch strategies retrospectively between calendar years or claim that observed annual differences establish a stable market regime rule.

The coefficient paths are also descriptive. Rolling estimates

Table 3. Calendar-year compounded net returns. The portfolio path is continuous across year boundaries.

Year	Ridge (%)	MA20 (%)	Buy-and-hold (%)
2022	-13.80	-17.99	-18.82
2023	16.15	19.93	27.09
2024	16.90	9.09	25.40
2025	9.88	7.67	17.79

can change as old outcomes leave the fixed window and new outcomes become available. Changes may reflect a shifting relation, correlated predictors, or estimation noise. They do not authorize removing a feature after seeing the test period. Full monthly fit boundaries and coefficient summaries are retained in the supplement so that this limitation is visible rather than hidden behind the aggregate result.

7 Discussion and Limitations

The experiment separates three claims that can otherwise be conflated. First, a pipeline can preserve time order and reproduce a numerical result. Second, a model can generate a portfolio with lower realized risk. Third, a strategy can deliver superior net investment performance. The evidence supports the first claim and documents a sample-specific risk reduction. It does not support the third. Methodological correctness is necessary for an interpretable negative finding; it is not evidence that the model must contain a profitable signal.

Selection and historical hindsight. The four-candidate grid is intentionally small, yet selecting by validation Sharpe can still favor estimation noise. The strong validation performance does not transfer to the holdout. The study cannot uniquely attribute this decline to a regime change, selection optimism, or weak underlying predictability. Moreover, the design was implemented after the test years had occurred. Fingerprints enforce consistency of the implemented sequence,

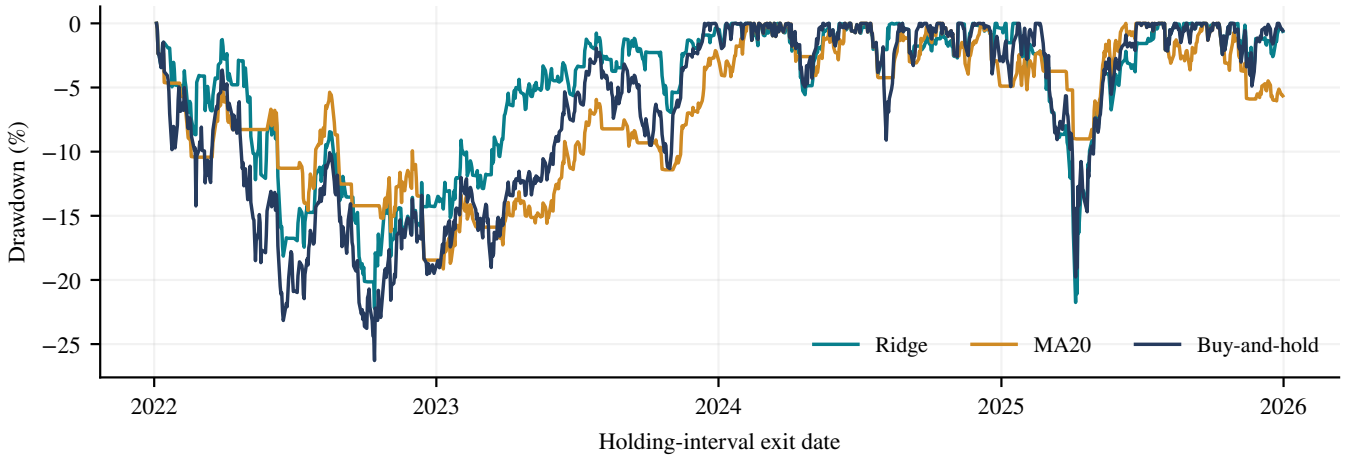


Figure 4. Net drawdowns, including initial wealth in the running peak. Lower drawdown is considered jointly with growth, Sharpe, and equity exposure. It is not a bound on future losses or evidence of successful crash prediction.

not independence from all prior familiarity with market history.

Execution and financing. Orders are assumed to fill at the next observed opening price, with fixed proportional costs. Auction participation constraints, depth, market impact, rejected orders, and partial fills are absent. Cash and the Sharpe reference rate are zero, and distributions are credited as ex-date receivables. These simplifying assumptions affect economic interpretation. The final comparison is internally consistent, but is not a deployment-ready simulation of a specific account.

Scope and statistical uncertainty. One ETF and one four-year holdout cannot establish broad generality. The reported Sharpe ratios are point estimates, and no dependence-robust confidence intervals, formal superiority tests, or factor alpha estimates are supplied. In particular, the small gross Sharpe difference is not presented as significant. Calendar-year breakdowns and diagnostic plots are descriptive; examining them does not increase the number of independent holdout observations.

Model limitations and future work. The model is additive and linear, its feature set is narrow, and monthly refitting can lag rapid changes. Binary allocations ignore confidence and expected transaction costs. A no-trade region, smoother position map, or pre-specified investable risk-matched benchmark is a reasonable future question. Each would alter the research design and require a new evaluation protocol. None is chosen retrospectively here as an allegedly improved strategy.

8 Conclusion

A small Ridge model with explicit information timing, validation-only selection, and reproducible accounting does not outperform SPY buy-and-hold after the stated costs. Its 6.54% CAGR and 0.481 Sharpe fall below the benchmark's 11.18% and 0.670. Lower drawdown accompanies lower equity exposure, while frequent switching turns approximate gross Sharpe parity into net underperformance. The study's useful output is a transparent empirical result and an inspectable explanation of the implementation gap. The accompanying supplement supplies the target derivation, accounting proofs, complete tables, fit audits, and exact reproduction commands.

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A Execution-Time Semantics and Corporate Actions

A.1 Three timestamps per observation

An observation has a decision date, an entry date, and an exit date. Treating these as interchangeable is a common source of hidden look-ahead. For a feature row indexed by session t, the decision occurs after close t, entry occurs at open $t + 1$, and the outcome is known at open $t + 2$. The model uses the signal-date index to retrieve features, the exit date to determine training eligibility, and entry/exit dates to assign holding returns to a reporting period. None of these operations can be replaced safely by assuming that an earlier row index implies an already realized outcome.

Table 4 gives the first six actual holdout observations. The first signal is formed on 31 December 2021 even though the first investment interval belongs wholly to 2022. On 3 January 2022, the model can be refitted using a label that ends at that morning's opening, but not using the label associated with the immediately preceding decision if it ends on 4 January. The code stores the maximum label exit in every fit audit so this distinction can be checked directly.

Table 4. Actual beginning-of-holdout trace. Forecasts and realized holding returns are in basis points. These are recorded outcomes, not a synthetic execution example.

Decision	Entry	Exit	Forecast (bps)	Realized (bps)	w
2021-12-31	2022-01-03	2022-01-04	4.832	61.306	1
2022-01-03	2022-01-04	2022-01-05	12.675	-42.986	1
2022-01-04	2022-01-05	2022-01-06	12.396	-194.274	1
2022-01-05	2022-01-06	2022-01-07	18.088	1.282	1
2022-01-06	2022-01-07	2022-01-10	21.408	-112.191	1
2022-01-07	2022-01-10	2022-01-11	15.164	54.679	1

The action chosen at the close is a target for the following opening auction. Until that opening, the previous position continues to bear the market movement of its own interval. At the next open the previous interval is marked, any new target is executed, and the next interval begins. This chronological decomposition ensures that the opening gap preceding entry is not credited to a position chosen only after the prior close. It also ensures that selling fees are assigned when the position changes, rather than appended to an unrelated later return.

A.2 Dividend entitlement and the target formula

Suppose a share is purchased at O_{t+1}. At exit its economic value consists of the sale price O_{t+2} and, if the exit is an ex-dividend date, the entitled distribution receivable D_{t+2}. The resulting gross wealth factor is

$$1 + y_t = \frac{O_{t+2} + D_{t+2}}{O_{t+1}}. \tag{A.1}$$

If $t + 1$ itself is an ex-date, a purchase at that open does not receive D_{t+1}. Adding that amount to the numerator would overstate performance. Conversely, omitting D_{t+2} when the position was carried into that ex-date would understate it. A unit test places a synthetic dividend on a known session and checks both cases independently.

This treatment is an ex-date total-return convention. It treats the receivable as reinvestable at the exit open; the actual payment date is not modeled. Across consecutive invested intervals, the convention compounds the dividend into portfolio wealth. A real account might instead carry a receivable or temporarily hold cash until payment. The convention is applied consistently to Ridge, MA20, and buy-and-hold, and is disclosed because internal consistency does not remove its economic approximation.

A.3 Raw opens and adjusted feature prices

The data source supplies OHLC and a separate adjusted close. Execution uses OHLC with explicit distributions, while features use trailing adjusted-price ratios. The absence of splits in this SPY interval means no split-share conversion is needed in the implemented P&L. The code refuses a split rather than assuming that vendor field semantics automatically solve every share-accounting problem. A different asset or extended sample containing splits would require an additional implementation and verification step before this target could be reused.

Vendor adjusted-close returns need not equal explicit cash-dividend returns at machine precision. For close C_t, define the explicit series

$$r_t^{\text{cash}} = \frac{C_t + D_t}{C_{t-1}} - 1, \qquad r_t^{\text{vendor}} = \frac{A_t}{A_{t-1}} - 1. \tag{A.2}$$

Their mean absolute difference in this snapshot is 0.00594 bps and their maximum absolute difference is 2.52010 bps. These differences reflect adjustment/reinvestment conventions and numerical representation. We report them rather than asserting exact equivalence. They do not enter execution twice: the portfolio return always follows Eq. (A.1).

Table 5. Representative dividend checks from the first and last events in the snapshot. Δ is vendor return minus explicit cash return. No representative split check is applicable because no split occurs.

Ex-date	Dividend (USD)	Vendor return (%)	Cash return (%)	Δ (bps)
2010-03-19	0.4800	-0.50618	-0.50410	-0.20801
2010-06-18	0.5310	0.10842	0.10790	0.05112
2025-09-19	1.8310	0.49528	0.49392	0.13685
2025-12-19	1.9930	0.90636	0.90367	0.26878

A.4 Calendar completeness and outliers

The observed 4,024 dates equal the expected XNYS session set over the observed endpoints. The check uses set differences in both directions, so a duplicate or accidental non-trading-day observation cannot cancel a missing session by preserving the row count. Special exchange closures are included in the calendar; the five-calendar-day gap ending 31 October 2012 therefore does not imply a missing trading observation. Half-days remain trading sessions and contribute one opening-to-opening interval when adjacent data are available.

Finite-price and OHLC ordering checks are separate from calendar verification. A complete calendar does not guarantee valid prices, and valid prices do not guarantee complete dates. Missing values cause failure rather than forward filling. Extreme observed returns are preserved because discarding crises after looking at the full history can alter the target distribution. No robust-scaling or clipping parameter is estimated from the full sample.

B Feature Invariance, Scaling, and Eligible Training Sets

B.1 Why trailing ratios do not use an absolute adjusted level

Future corporate actions can rescale earlier vendor-adjusted prices. The relevant question is whether that rescaling changes the information used by an earlier predictor, not whether the downloaded file was created at the historical decision time. For the particular features used here, a common factor across the required lookback cancels algebraically.

Proposition 2 (Common-factor invariance)**.** *Fix a decision session t. Suppose an adjustment replaces A_s by $\widetilde{A}_s = \lambda A_s$ for all $s \in \{t-20, \dots, t\}$, where $\lambda > 0$. The five-dimensional feature vector defined in Eqs. (3)–(5) is unchanged.*

Proof. For each $k \in \{1, 5, 20\}$,

$$\frac{\widetilde{A}_t}{\widetilde{A}_{t-k}} - 1 = \frac{\lambda A_t}{\lambda A_{t-k}} - 1 = \frac{A_t}{A_{t-k}} - 1. \tag{B.3}$$

Similarly, the denominator of the moving-average ratio is multiplied by λ, so $\widetilde{g}_t = g_t$. Every one-session return needed for v_t has numerator and denominator in the stated window; each ratio is unchanged. The vector of 20 returns, its sample mean, and its sample standard deviation are therefore unchanged. □

The proposition is deliberately narrow. It does not cover arbitrary data corrections, changing field definitions, or a transformation that is nonconstant within the relevant earlier lookback. Nor does it license future corporate-action quantities as independent predictive inputs. The source snapshot and digest remain necessary. A regression test multiplies all adjusted closes through a cutoff by a common factor and checks equality of all earlier features up to numerical tolerance.

B.2 Exact lookbacks and missing-value policy

The 20-session return uses A_t and A_{t-20}. The moving average uses 20 levels from A_{t-19} through A_t. Volatility uses the 20 returns ending on those dates and consequently also requires A_{t-20}. All five features become simultaneously available after 20 warm-up sessions. Using a partial initial rolling window would create a different feature definition and is not allowed silently.

The raw-history range extends beyond the first possible trading observation for this reason. The full history contains 4,024 sessions; deleting 20 warm-up rows and two terminal unknown labels leaves 4,002 complete samples. Validation and testing select their periods from that common sample using holding dates. Other missing data would require inspection and a documented source correction, not automatic interpolation that might use later information.

B.3 Training-only standardization

At refit d, for feature coordinate j and $n_d = |\mathcal{T}_d|$,

$$\mu_{d,j} = \frac{1}{n_d} \sum_{t \in \mathcal{T}_d} x_{t,j}, \tag{B.4}$$

$$s^2_{d,j} = \frac{1}{n_d} \sum_{t \in \mathcal{T}_d} (x_{t,j} - \mu_{d,j})^2. \tag{B.5}$$

This denominator is n_d, matching `StandardScaler`; it differs from the denominator 19 used to define the 20-return volatility feature. Confusing the two standard deviations would not be a timing leak, but would change the numerical penalty interpretation and prevent exact reconstruction. The scaler is fitted with the model on the same eligible rows and then remains fixed until the next refit.

Table 6. Recorded training means and scales at the first and last holdout fits. Values use decimal-return units, not percentage points. These differ across windows and are not estimated from a full-sample fit.

Feature	First mean	First scale	Last mean	Last scale
`return_1`	0.000734	0.011871	0.000623	0.010769
`return_5`	0.003582	0.023666	0.002992	0.022403
`return_20`	0.014244	0.046303	0.012595	0.041769
`ma_gap_20`	0.006368	0.026229	0.005504	0.023464
`volatility_20`	0.009079	0.007716	0.009757	0.004768

B.4 Observable labels and scheduled updating

For a decision close d, define $\mathcal{E}_d = \{t : t < d,\ e(t) \le d\}$. The implemented training set is its latest 1,260 rows. Because each included outcome ends at or before open d, and all its features are available by close $t < d$, the entire training pair is observable by close d. Outcomes ending later than d are excluded even if the feature row is already known. The filter is equivalent to the appropriate non-overlap rule for this one-interval target; an arbitrary additional embargo is not needed to make already mature, non-overlapping labels observable.

This reasoning would change for a longer holding horizon, delayed fundamental announcements, revised macroeconomic vintages, or overlapping labels used in validation. In those settings, timestamping a row by calendar date would not establish its publication time, and a revised maturity or purging rule would be required. The current project avoids those extra information sources rather than claiming a general solution to every financial-data timing problem.

Monthly refitting updates model parameters under a fixed policy. There is no requirement that all training samples precede the beginning of the entire holdout: once an earlier test outcome occurs, it becomes legitimate past information for a later decision. What remains locked is the procedure that maps available data to predictions, including features, window, refit schedule, penalty, and position rule. Reselecting these choices using holdout outcomes would be a separate and invalid use of the final evaluation.

B.5 Objective scaling and coefficient interpretation

For centered standardized features Z and centered outcomes y, Ridge has the solution

$$\widehat{\beta} = (Z^\top Z + \alpha I)^{-1} Z^\top y, \tag{B.6}$$

under the stated sum-loss convention. The implementation uses SVD rather than explicitly forming the inverse. With standardized features, coefficients represent decimal-return changes for a one-training-SD feature change. Multiplying a coefficient by 10^4 expresses that conditional association in basis points. It is not an elasticity and not the effect of a one-percentage-point raw feature change.

The penalty reduces sensitivity to near-collinearity, but a ridge coefficient may still change sign as the window changes. The five technical signals share price history and overlap in their lookbacks. A sign on one coordinate is conditional on the others and cannot be interpreted independently as evidence for a structural momentum or reversal mechanism. Appendix E retains both paths and descriptive summaries without using them to prune the specification.

C Portfolio Accounting and Cost Attribution

C.1 From decisions to a self-consistent interval path

For each interval i, the portfolio first moves from the previous target w_{i-1} to the new target w_i at the entry open. Its fee is a fraction cq_i of pre-trade wealth, where $q_i = |w_i - w_{i-1}|$. The remaining wealth then earns $w_i y_i$, with the uninvested part earning zero. At the final exit, an invested portfolio liquidates and pays the additional cost. The interval and terminal factors are

$$G_i(c) = (1 - cq_i)(1 + w_i y_i), \tag{C.7}$$

$$V_N(c) = \left[\prod_{i=1}^{N} G_i(c)\right](1 - cw_N). \tag{C.8}$$

The assumption is a pre-trade-equity haircut. If a broker charges a fee on purchased notional while imposing exact full-investment constraints, the share quantity can require a slightly different cash equation. Equation (C.7) is therefore an explicit research convention. It is not silently described as a detailed brokerage implementation. All strategies share the convention.

Binary weights simplify the bookkeeping. An unchanged fully invested portfolio remains fully invested after its asset return; an unchanged cash portfolio remains cash. There is no fractional target drift to rebalance. Dividends are incorporated by the declared total-return convention. A continuously rebalanced fractional-weight benchmark would need its own cash and turnover treatment; it cannot be produced by simply copying this binary-trade count.

C.2 Proof and interpretation of the exact drag identity

Proof of Proposition 1. For fixed positions and returns, every market-return factor $(1 + w_i y_i)$ appears in both $V_N(c)$ and $V_N(0)$ and cancels. Since $w_i \in \{0, 1\}$, each q_i is also zero or one, and $(1 - cq_i) = (1 - c)^{q_i}$. The terminal factor is $(1 - c)^{w_N}$. Hence

$$\frac{V_N(c)}{V_N(0)} = \left[\prod_{i=1}^{N}(1 - c)^{q_i}\right](1 - c)^{w_N} \tag{C.9}$$

$$= (1 - c)^{\sum_i q_i + w_N} = (1 - c)^K. \tag{C.10}$$

All factors are positive under the proposition's conditions, so the ratio is well defined. □

Taking logarithms gives an exact annualized log-wealth drag,

$$L(c) = -\frac{252}{N}\log\frac{V_N(c)}{V_N(0)} = -\frac{252K}{N}\log(1 - c). \tag{C.11}$$

For small c, $-\log(1 - c) \approx c$, producing the intuitive first-order relationship $L(c) \approx 252Kc/N$. The numerical comparison in Table 7 uses the exact formula, not the approximation. The builder checks the identity against the saved net and gross terminal wealth values before emitting the table.

Table 7. Exact fee attribution at 5 bps, conditional on the same frozen positions. Annual log drag differs from a difference in arithmetic returns or percentage-point CAGR.

Strategy	K	Net/gross wealth	Annual log drag (%)
Ridge	246	0.884236	3.094
MA20	102	0.950267	1.283
Buy-and-hold	2	0.999000	0.025

The identity shows how much wealth is removed by the declared fees, but does not by itself explain why the gross strategy differs from buy-and-hold. That difference reflects which returns are held and which are missed. It also does not identify a counterfactual strategy that would trade less: changing the threshold would change w_i, the gross market factors, and K together. Reporting a lower-cost threshold without a separate evaluation would mix accounting attribution with post-hoc strategy selection.

C.3 A small synthetic accounting example

Consider three hypothetical holding returns $(0.01, -0.02, 0.03)$ and target weights $(1, 0, 1)$. These numbers illustrate the formula only and do not replace any empirical return. Beginning from cash, entry/switch turnover is $(1, 1, 1)$, and final liquidation adds one more unit. With cost c, terminal wealth is

$$V_3(c) = [(1 - c)1.01]\,[(1 - c)]\,[(1 - c)1.03]\,(1 - c) = (1 - c)^4(1.01)(1.03). \tag{C.12}$$

The negative middle asset return is not earned because the portfolio is in cash, but the sale into cash still incurs a fee. An always-cash path earns one with no fees; an always-invested path incurs exactly two fee units, at entry and exit. The integrity suite checks all three cases. Charging fees only on purchases, omitting final liquidation, or charging a daily fee on unchanged positions would produce a different and incorrect implementation of the declared rule.

C.4 Metric definitions and calendar aggregation

Initial capital is one. Terminal wealth determines total return and CAGR. Annualized arithmetic return and volatility use 252 intervals and sample standard deviation; Sharpe divides those quantities with a zero reference rate. Maximum drawdown is

$$\mathrm{MDD} = \min_i \left(\frac{V_i}{\max(1, V_1, \ldots, V_i)} - 1 \right). \tag{C.13}$$

Including initial capital is necessary: otherwise a loss in the first interval would be excluded from the drawdown history. The implementation has a dedicated regression test for this case.

Calendar-year returns compound only that year's continuous-path net interval returns. Costs are not reset on 1 January. The initial fee belongs to the first interval and final liquidation belongs to the last interval; intervening year boundaries create no trades. Annual sample counts differ because market calendars differ. Annual returns in the supplement are realized compounded returns, while an annual row's CAGR may use $252/N_{\mathrm{year}}$ and therefore differ slightly. The main annual table intentionally uses realized return rather than relabeling it as a full-period CAGR.

D Complete Empirical Results and Frozen-Path Diagnostics

D.1 Complete validation search

Table 8 includes all four candidates, not only the selected row. The selection criterion is net Sharpe at 5 bps. The forecast R^2 columns are descriptive and are not a second objective used to resolve the selected outcome. Penalties 0.1 and 1 have identical validation trading results but different forecast errors because the binary position map can hide changes in magnitude that do not cross zero.

Table 8. Full pre-declared validation grid. CAGR, volatility, and drawdown are in percent; turnover is the total charged one-way amount. Validation contains 756 holding intervals per candidate.

α	CAGR (%)	Sharpe	Vol. (%)	MDD (%)	Turnover	R^2_{OOS}
0.1	22.33	1.2060	18.08	-32.05	136	-0.01442
1	22.33	1.2060	18.08	-32.05	136	-0.01434
10	19.15	1.0767	17.76	-32.05	134	-0.01364
100	18.64	1.0310	18.21	-32.05	116	-0.01089

Validation returns are not independent observations from the final test, and overlapping rolling training windows create dependence among successive forecasts. The selected validation Sharpe is consequently a development statistic, not a clean estimate of deployable performance. The experiment does not interpret the gap between validation and testing as a uniquely identified structural break. Multiple explanations, including sampling noise and selection optimism, remain consistent with the evidence.

D.2 All cost scenarios and annual outcomes

Table 9 reports all three strategies at zero, five, and ten basis points. Positions and predictions are fixed across scenarios. The passive benchmark's near-flat cost curve reflects its two charged trades; MA20's intermediate sensitivity reflects its 102 trades; Ridge is most sensitive with 246. These comparisons would be obscured if costs were waived for one strategy or if each scenario were allowed to reselect a different model.

The annual table emphasizes that a lower full-period maximum drawdown need not mean consistently better returns. Ridge loses less than buy-and-hold in 2022, but its realized return is lower in the following three years. A rule that alternated between these strategies after seeing the annual table would be a new, hindsight-based portfolio rather than the evaluated experiment. No such switching rule is included.

D.3 Forecast distribution and direction classification

Forecast MSE is 0.0001336494, compared with 0.0001329298 for the past-only historical mean. Their ratio yields $R^2_{\mathrm{OOS}} = -0.00541$. All errors refer to the executable label, not a more convenient close-to-close target. Table 11 reports the scale of

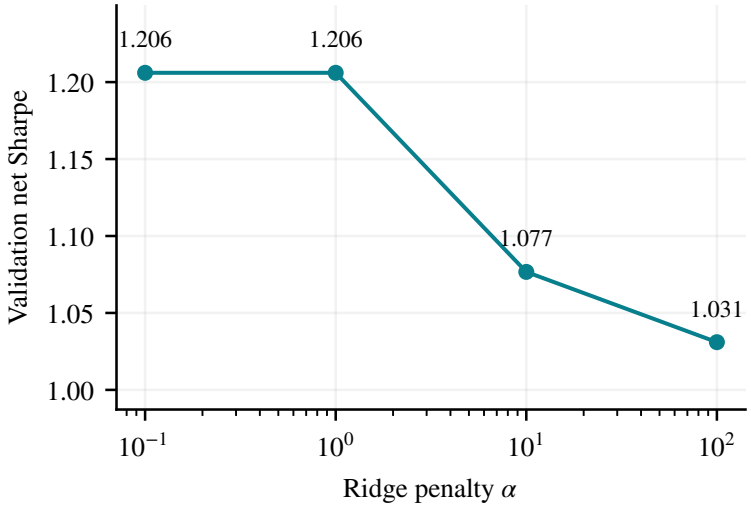


Figure 5. Validation penalty path. Only these four settings are compared. There is no corresponding holdout penalty curve because the final period is reserved for the selected setting.

Table 9. Full cost sensitivity of the same frozen holding paths. Final wealth includes liquidation. Cost values are assumptions rather than estimates of actual opening-auction spreads or impact.

Strategy	bps	CAGR (%)	Sharpe	MDD (%)	Final wealth
Ridge	0	9.88	0.678	-20.97	1.4546
MA20	0	5.03	0.508	-17.84	1.2156
Buy-and-hold	0	11.20	0.672	-26.28	1.5254
Ridge	5	6.54	0.481	-22.20	1.2863
MA20	5	3.69	0.389	-19.14	1.1552
Buy-and-hold	5	11.18	0.670	-26.28	1.5238
Ridge	10	3.29	0.284	-23.89	1.1373
MA20	10	2.37	0.271	-20.43	1.0977
Buy-and-hold	10	11.15	0.669	-26.28	1.5223

Table 10. Full calendar-year net results at 5 bps. Return, volatility, and exposure are in percent. Turnover is assigned to the intervals where the continuous portfolio actually trades.

Strategy	Year	Return (%)	Vol. (%)	Sharpe	Exposure (%)	Turnover
Ridge	2022	-13.80	20.11	-0.644	65.60	60
Ridge	2023	16.15	10.60	1.478	67.20	69
Ridge	2024	16.90	9.47	1.696	73.02	59
Ridge	2025	9.88	19.59	0.582	71.60	58
MA20	2022	-17.99	13.28	-1.439	42.00	32
MA20	2023	19.93	9.61	1.954	67.20	21
MA20	2024	9.09	9.18	0.993	80.16	23
MA20	2025	7.67	10.69	0.751	71.60	26
Buy-and-hold	2022	-18.82	24.15	-0.750	100.00	1
Buy-and-hold	2023	27.09	13.47	1.861	100.00	0
Buy-and-hold	2024	25.40	12.65	1.854	100.00	0
Buy-and-hold	2025	17.79	20.41	0.910	100.00	1

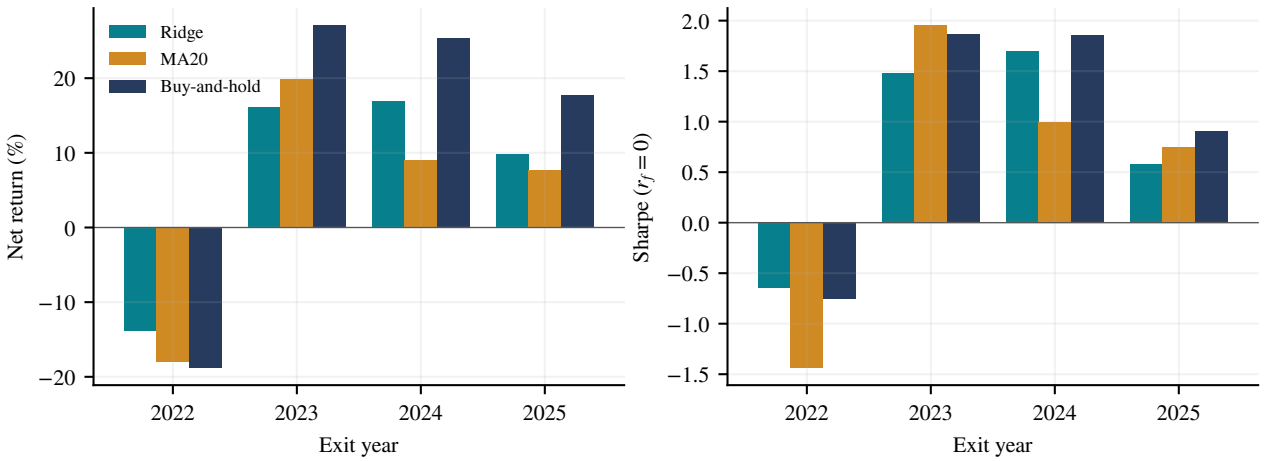


Figure 6. Annual net return and Sharpe. These panels describe the four observed years; they do not provide four independent replications of a strategy-selection experiment.

forecasts and outcomes in basis points; the forecast distribution is much narrower than the realized-return distribution.

Table 11. Descriptive distribution of the frozen holdout forecasts and outcomes. SD is the sample standard deviation. These statistics are not used for full-sample scaling or threshold selection.

Series (bps)	Mean	SD	Min.	Median	Max.
Model forecast	5.867	12.797	-46.704	4.091	73.544
Realized holding return	4.879	115.328	-658.430	8.726	784.897
Historical-mean forecast	5.852	0.842	4.271	6.160	7.590

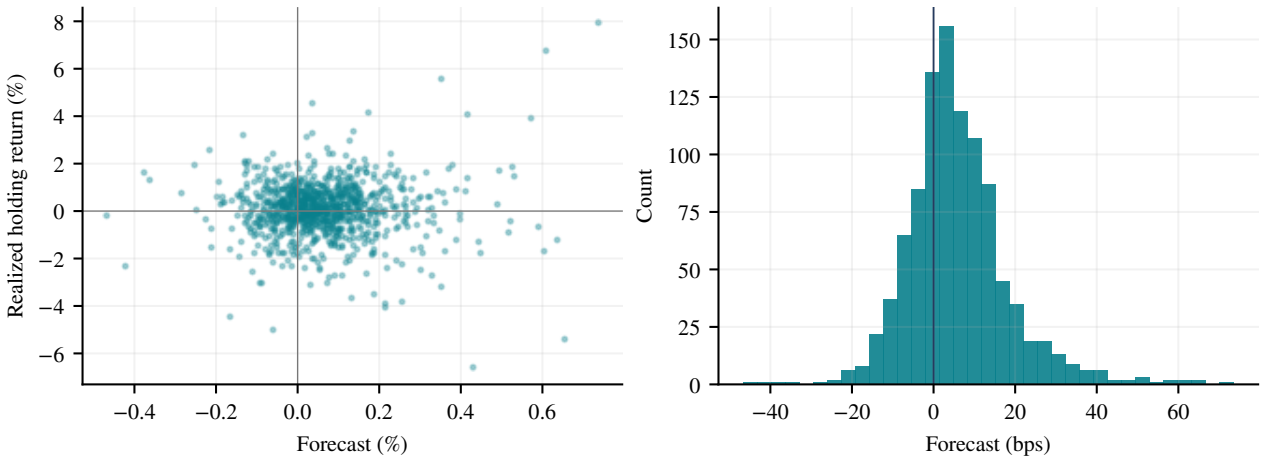


Figure 7. Frozen forecasts against realized returns, and the distribution of forecast scores. The zero line is the pre-specified investment threshold, not a threshold fitted to this plot.

The confusion matrix makes clear why a hit rate is not an economic objective. A correct decision on a tiny return counts as much as a correct decision on a large return. Predicting positive returns frequently may also achieve a favorable hit rate simply when the unconditional positive-return fraction is high. The matrix is therefore interpreted alongside magnitude, exposure, turnover, and net wealth. It is not used to recalibrate a probability threshold or to claim a classification advantage.

Conditional returns are a descriptive partition of the single test path. They are neither randomized treatment effects nor independent samples, and differences can be affected by the dates at which the model is invested. We do not turn this partition into an additional significance claim. Its purpose is to connect the model's action to the asset returns that were held or missed, while keeping fees in the separate portfolio accounting.

Table 12. Direction classification and position map. Positive forecasts imply SPY; all other forecasts imply cash. Counts use the realized asset return sign, not the portfolio return after costs.

Forecast position	$y > 0$	$y \leq 0$	Total
SPY	371	324	695
Cash	165	142	307

Table 13. Underlying realized holding returns conditional on Ridge's frozen position. The cash row reports returns that were not earned; it does not assume a hypothetical cash strategy earns those asset returns.

Ridge decision	Count	Mean y (bps)	SD (bps)	Positive (%)
Invested intervals	695	6.097	118.821	53.38
Cash intervals	307	2.121	107.137	53.75

D.4 Exposure and the ex post diagnostic

Figure 8 shows monthly mean exposure. These averages are summaries of binary daily positions, not fractional weights used in execution. The ex post constant-exposure diagnostic multiplies the underlying holding return by the realized average Ridge exposure, approximately 0.6936, and ignores fractional rebalancing costs. It has gross CAGR 8.03%, volatility 12.70%, and Sharpe 0.672 in this path. These values are not included in the investable benchmark table because the exposure was learned from the holdout.

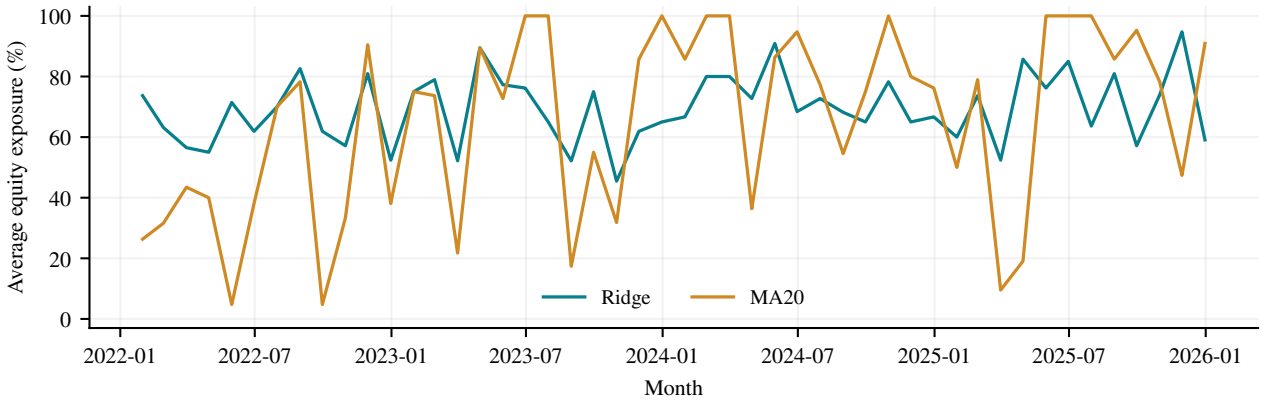


Figure 8. Monthly average equity exposure. Time in cash reduces participation in both losses and gains. The visual pattern is a diagnostic of the frozen decision rule, not an additional market-state model.

An investable risk-matched comparison would choose its risk budget using information available before the evaluated interval. It would also specify how weights respond to changes in wealth and how rebalancing costs are charged. That construction is outside this minimal experiment. Keeping the current series labeled ex post prevents an appealing diagnostic from being misrepresented as a strategy that could have been selected in advance.

E Monthly Fits and Parameter Audit

The final holdout contains 49 fits. The first occurs at the 31 December 2021 decision close; the first decision in January creates the next scheduled fit, followed by each subsequent month through December 2025. Every fit has 1,260 training observations. The complete boundaries are retained in Table 14. The latest label exit never exceeds the fit date. The last training signal can be earlier than the previous exchange session because of the two-session target offset.

The boundary table is a useful audit but does not, by itself, prove that every processing stage is past-only. For that reason it is supplemented by tests that compare actual scaler means with the eligible-window means and by future-perturbation tests that check predictions and audit records. The combination of an explicit timing rule, exported boundaries, and behavioral tests is stronger than relying on an assertion of rolling estimation alone.

Coefficient changes can occur even with a stable data-generating relation because old observations leave the fixed-length window and new observations enter. Correlated predictors can redistribute weight among themselves. A large coefficient on the moving-average gap, for example, is not evidence that the corresponding univariate rule should replace the full model. The relation is estimated jointly, after scaling, and is assessed only through the frozen out-of-sample procedure. No coefficient-stability filter is added to improve the reported path.

Table 14. All final-holdout fit boundaries. The latest label exit is already observed at the fit-date open. A fixed training sample count does not imply a fixed calendar span because exchange-session calendars differ across years.

Fit date	First training signal	Last training signal	Latest label exit	N
2021-12-31	2016-12-28	2021-12-29	2021-12-31	1260
2022-01-03	2016-12-29	2021-12-30	2022-01-03	1260
2022-02-01	2017-01-30	2022-01-28	2022-02-01	1260
2022-03-01	2017-02-27	2022-02-25	2022-03-01	1260
2022-04-01	2017-03-30	2022-03-30	2022-04-01	1260
2022-05-02	2017-04-28	2022-04-28	2022-05-02	1260
2022-06-01	2017-05-30	2022-05-27	2022-06-01	1260
2022-07-01	2017-06-28	2022-06-29	2022-07-01	1260
2022-08-01	2017-07-27	2022-07-28	2022-08-01	1260
2022-09-01	2017-08-29	2022-08-30	2022-09-01	1260
2022-10-03	2017-09-28	2022-09-29	2022-10-03	1260
2022-11-01	2017-10-27	2022-10-28	2022-11-01	1260
2022-12-01	2017-11-28	2022-11-29	2022-12-01	1260
2023-01-03	2017-12-28	2022-12-29	2023-01-03	1260
2023-02-01	2018-01-29	2023-01-30	2023-02-01	1260
2023-03-01	2018-02-26	2023-02-27	2023-03-01	1260
2023-04-03	2018-03-29	2023-03-30	2023-04-03	1260
2023-05-01	2018-04-26	2023-04-27	2023-05-01	1260
2023-06-01	2018-05-29	2023-05-30	2023-06-01	1260
2023-07-03	2018-06-27	2023-06-29	2023-07-03	1260
2023-08-01	2018-07-26	2023-07-28	2023-08-01	1260
2023-09-01	2018-08-28	2023-08-30	2023-09-01	1260
2023-10-02	2018-09-26	2023-09-28	2023-10-02	1260
2023-11-01	2018-10-26	2023-10-30	2023-11-01	1260
2023-12-01	2018-11-27	2023-11-29	2023-12-01	1260
2024-01-02	2018-12-27	2023-12-28	2024-01-02	1260
2024-02-01	2019-01-29	2024-01-30	2024-02-01	1260
2024-03-01	2019-02-27	2024-02-28	2024-03-01	1260
2024-04-01	2019-03-27	2024-03-27	2024-04-01	1260
2024-05-01	2019-04-29	2024-04-29	2024-05-01	1260
2024-06-03	2019-05-30	2024-05-30	2024-06-03	1260
2024-07-01	2019-06-26	2024-06-27	2024-07-01	1260
2024-08-01	2019-07-29	2024-07-30	2024-08-01	1260
2024-09-03	2019-08-28	2024-08-29	2024-09-03	1260
2024-10-01	2019-09-26	2024-09-27	2024-10-01	1260
2024-11-01	2019-10-29	2024-10-30	2024-11-01	1260
2024-12-02	2019-11-26	2024-11-27	2024-12-02	1260
2025-01-02	2019-12-27	2024-12-30	2025-01-02	1260
2025-02-03	2020-01-28	2025-01-30	2025-02-03	1260
2025-03-03	2020-02-25	2025-02-27	2025-03-03	1260
2025-04-01	2020-03-25	2025-03-28	2025-04-01	1260
2025-05-01	2020-04-24	2025-04-29	2025-05-01	1260
2025-06-02	2020-05-26	2025-05-29	2025-06-02	1260
2025-07-01	2020-06-23	2025-06-27	2025-07-01	1260
2025-08-01	2020-07-24	2025-07-30	2025-08-01	1260
2025-09-02	2020-08-24	2025-08-28	2025-09-02	1260
2025-10-01	2020-09-23	2025-09-29	2025-10-01	1260
2025-11-03	2020-10-26	2025-10-30	2025-11-03	1260
2025-12-01	2020-11-20	2025-11-26	2025-12-01	1260

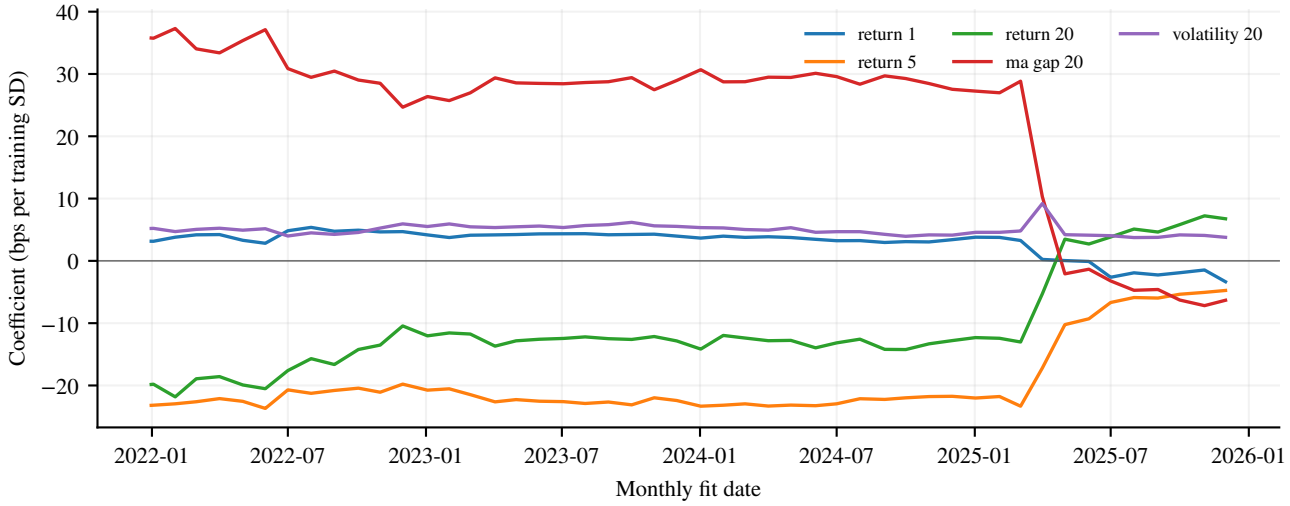


Figure 9. Coefficient paths on training-standardized inputs, in basis points per training SD. Signs and magnitudes are conditional associations among correlated features. The paths are not used to select features after testing.

Table 15. Descriptive coefficient summary across the 49 fitted models. Sign changes count adjacent monthly estimates with different signs, not changes in the daily position.

Feature	Mean (bps)	Min. (bps)	Max. (bps)	Sign changes
`return_1`	2.918	-3.353	5.376	1
`return_5`	-19.585	-23.684	-4.748	0
`return_20`	-10.991	-21.827	7.227	1
`ma_gap_20`	23.807	-7.187	37.297	1
`volatility_20`	4.959	3.742	9.203	0

F Reproduction, Artifact Integrity, and Team Metadata

F.1 Environment and exact stage commands

Python 3.12 is the tested interpreter. Direct dependency versions are pinned in `requirements.txt`; a clean-install transitive environment is recorded in `requirements-lock.txt`. The seed is 42, while the Ridge SVD solve itself is deterministic for the stated inputs. Numerical libraries and platform-specific rendering can affect bytes, so input and result digests are reported separately from a claim of universal bitwise equality.

From a clean checkout, the recommended commands preserve published outputs and create a separate reproduction directory:

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m pytest -q
python -m project2.pipeline all --workspace runs/reproduction
python -m project2.report --workspace runs/reproduction
```

The pipeline command downloads history only if its local cache is absent, checks data, builds features, evaluates four validation candidates, locks the selection, evaluates only the chosen holdout model, and writes tables and plots. The report command reads saved results, generates the paper's tables and vector figures, and compiles this LaTeX document. It does not fit a new trading model. Tectonic or a standard LaTeX installation is required for typesetting; that requirement is separate from Python packages.

Independent stage commands are available for inspection. A root-directory run is appropriate only when working with the published matching input snapshot, or when intentionally maintaining a separate documented experiment. Appending `--workspace runs/reproduction` to each pipeline stage uses the separate directory consistently.

```
python -m project2.pipeline fetch
python -m project2.pipeline validate
python -m project2.pipeline test
python -m project2.pipeline figures
python -m project2.report
```

F.2 Artifact map and preserved input identity

Table 16. Source-to-artifact map. Paths are repository-relative; no local absolute machine directories are required by the research pipeline.

Artifact	Role
`config.json`	Frozen research specification and separate, editable identity fields.
`data/provenance.json`	Source options, retrieval metadata, observed coverage, and raw-file digest.
`output/selection.json`	Selected penalty and protocol/input/validation fingerprints.
`output/tables/test_predictions.csv`	Signals, holding dates, forecasts, outcomes, and positions.
`output/tables/test_fit_audit.csv`	Eligible-window boundaries, means, scales, intercepts, and coefficients.
`output/tables/test_path_*.csv`	Per-interval returns, fees, turnover, and compounded wealth.
`output/run_manifest.json`	Original research environment, settings, sample dates, and result digests.
`paper/main.tex`	Conference-style report source, with modular sections and bibliography.
`paper/generated/`	Saved-result tables, vector figures, and source-artifact digests.

The following SHA-256 identities distinguish the original input snapshot from the final prediction artifact:

Artifact	SHA-256
Raw input	`e59594912552818d0fce625e8c8f16efe49d88db067cd787c27995355aa46498`
Predictions	`0ccaeaf2351cf3c005bced9b61d1a2db97632c90aa4ba91c39a32692d81aaa2c`

The raw price file is not redistributed. A fresh vendor download may change after history revisions or adjustment rounding. Exact result reproduction is conditional on a matching input snapshot; different input is a newly disclosed run. Within an established run, a changed protocol or raw-data digest causes the selection lock to refuse overwriting the original experiment.

F.3 Integrity checks and clean-environment evidence

The suite contains 11 tests covering: executable label alignment and dividend entitlement; future-price invariance of earlier features; common-factor adjustment invariance; mature-label eligibility; invariance of earlier forecasts and scaling to future changes; equality of recorded means with training-window means; monthly refit count; initial and terminal fees; cash and switching behavior; initial-capital drawdown; period-boundary allocation; and explicit failure on missing values and splits. Some tests cover multiple related cases. Synthetic fixtures exist only in this suite, never as a fallback for reported performance.

A clean-environment reproduction used a fresh isolated virtual environment and a fresh working directory with the same downloaded local input snapshot. The validation-grid, final-prediction, fit-audit, and headline-metric CSVs matched byte for byte. This verifies reproducibility conditional on that input; it is not a second independent market sample. The recorded evidence is in `output/reproducibility_check.json`. Hosted continuous integration checks the suite and rebuilds the report from published derived results without downloading changing market data.

The LaTeX tables and numeric macros are generated from the same frozen CSV/JSON results, and the typesetting build verifies that its source digests match those artifacts. Rewriting this report does not refit the trading strategy or alter the selected penalty. This separation matters because presentation improvements should not silently become a new model-selection exercise.

F.4 Assumptions a reviewer should retain

All results depend on next-open fills, a proportional pre-trade fee, zero cash interest, a zero Sharpe reference rate, and ex-date receivable crediting. There is no order-book depth or market impact model, no funding or borrow constraint because the strategy is unlevered and long-only, and no split event in the supported sample. Full-history adjusted-price ratios have the stated common-factor invariance but remain subject to arbitrary vendor corrections. The single-asset and four-year-holdout scope is narrow, and no formal superiority test or factor alpha is claimed.

F.5 Team information and assistance disclosure

Team identity and contribution fields are intentionally blank and excluded from the protocol digest; filling them does not change model selection. Before submission, add the actual group number, title-page roster, and genuine contributions, consistently with the Canvas group.

Table 17. Team and contribution information, intentionally unfilled.

Group / team name	
Full member roster	
Member	Contribution

Code and report drafting used AI assistance; all reported results come from the executed historical-data pipeline. The walk-through uses disclosed synthetic narration over actual results and command outputs. Team members should review the artifacts, record true contributions, and follow course disclosure rules. No live trading is performed.